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On the Applicability of LLMs and SLMs for Privacy-Preserving Named Entity Recognition in Financial Applications

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Abstract

This work explores how deep learning models, with different numbers of parameters, can be effectively applied to detect personal data within unstructured text using Named Entity Recognition (NER) techniques. We evaluate the performance of various architectures by leveraging a plethora of language models (LMs) consisting of Distilbert-base-uncased, Distilbert-base-cased, Bert-base-uncased, Bert-base-cased, Bert-large-uncased, Bert-large-cased, ModernBERT-base, ModernBERT-large, nomic-BERT-2048, RoBERTa-base, DistilRoBERTa-base, RoBERTa-large, Deberta-v3-xsmall, Deberta-v3-small, and Deberta-v3-base, which are evaluated using the performance indices of accuracy, precision, recall, and F1-score. Our experiments show that some Small Language Models (SLMs) compete equally with some corresponding LLMs (Large Language Models), based on the specific PII (Personally Identifiable Information) dataset, thus enhancing personal data detection, which is of paramount importance in financial applications. Moreover, we proposed a novel architecture based on an optimized transformer fine-tuning strategy to improve PII recognition across diverse contexts and conducted an extensive comparative analysis to evaluate the performance of our proposed architecture in relation to all relevant existing approaches reported in the literature. This evaluation, performed on the AI4Privacy PII 43 K dataset, encompasses every publicly available work we identified and provides a thorough benchmarking of our methods within the current research field. The results highlight both the strengths and limitations of existing solutions and demonstrate the effectiveness of SLMs in addressing the challenges of privacy-preserving information extraction.

Keywords: small language models; large language models; named entity recognition; machine learning; financial application; privacy-preserving application

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1. Introduction

The rapid progress in Large Language Models (LLMs) has created opportunities in many domains, including cybersecurity, where the threat landscape continues to evolve, and organizations increasingly seek advanced defensive technologies [1]. Cybersecurity remains a pressing concern, as the frequency and sophistication of cyberattacks continue to grow, posing substantial risks to individuals, industry, and governmental infrastructures [2]. Recent advances in artificial intelligence, and LLMs in particular, have shown considerable promise in reshaping how cybersecurity challenges are addressed [3].

As demonstrated by Yigit et al. [3], modern LLMs can support a wide spectrum of security-related tasks such as threat intelligence analysis, vulnerability assessment, malware classification, anomaly detection, process automation, error correction, and even the generation of secure or insecure code depending on the prompts used. Research has also explored using LLMs to better understand adversarial behaviors. For example, Fayyazi et al. [4] examined how LLMs can interpret and summarize cyberattack Tactics, Techniques, and Procedures (TTPs) from the MITRE ATT&CK framework. Their findings show that Retrieval Augmented Generation (RAG) substantially improves the clarity and completeness of TTP explanations by injecting relevant contextual knowledge into the model's responses, underscoring LLMs' value for enhancing threat intelligence workflows.

Other studies leverage external knowledge bases through RAG to support vulnerability analysis. Du et al. [5] introduced Vul-RAG, a method that integrates knowledge-level retrieval with LLM reasoning to assist in vulnerability detection. Beyond improving detection accuracy, the vulnerability descriptions produced by Vul-RAG offer high-quality interpretability that can aid human analysts during manual assessments. LLMs have also been used as autonomous agents for complex defensive tasks. Rigaki et al. [6] demonstrated that LLM-driven agents can operate effectively within cybersecurity environments, achieving performance that rivals or surpasses that of heavily trained agents on sequential decision-making tasks, even without additional fine-tuning. Their work also introduced NetSecGame, a flexible and modular environment designed to support sophisticated multi-agent cybersecurity experimentation.

Belcak et al. [7] argue that Small Language Models (SLMs) offer a compelling alternative to large-scale models for many agentic AI scenarios. Their work suggests that SLMs provide sufficient capability for the language processing tasks typical in agent-based applications while also being operationally better suited to such settings due to their reduced computational footprint. The authors further emphasize that, because of their compact size, SLMs deliver these advantages at substantially lower cost, making them a practical and economical choice for the majority of agentic system use cases when compared to general-purpose LLMs. Similarly, Thamm et al. [8] note a rising preference for SLMs within application domains where latency, efficiency, and customizability are critical. They highlight that SLMs not only support faster inference and reduced resource consumption but also simplify fine-tuning and domain adaptation. These characteristics make them particularly appealing for environments with limited computational capacity or strict privacy requirements. The authors conclude that SLMs are well positioned for tasks involving localized data processing, ultra-low-latency decision-making, and targeted knowledge acquisition, areas where large models frequently encounter significant operational limitations.

Within Natural Language Processing (NLP), Named Entity Recognition (NER) has long been recognized as a core technique for transforming raw, unstructured text into meaningful structured information. By identifying and categorizing entities, such as people, organizations, locations, temporal expressions, and other domain-specific terms, NER enables more efficient information retrieval, downstream analysis, and knowledge extraction [9]. Its importance stems from its ability to impose structure on free-form text, making large corpora more interpretable and actionable.

In the financial domain, NER plays a particularly critical role due to the high volume and complexity of unstructured textual data generated by sources such as financial reports, earnings calls, regulatory filings, and news articles. Accurate identification of entities, including companies, financial instruments, monetary values, events, and temporal references, supports key tasks such as risk assessment, market analysis, compliance monitoring, and automated decision-making. By structuring financial text, NER enables more precise extraction of actionable insights and facilitates the integration of textual information into quantitative financial models. According to Zang [10], financial institutions process millions of documents annually, ranging from loan applications to regulatory filings, each containing critical structured information embedded within unstructured text. The automated extraction of entities such as organization names, monetary values, and regulatory identifiers remains computationally intensive and accuracy-critical. Peeters et al. [11] introduced a NER system built on a Bidirectional Encoder Representations from Transformers (BERT) architecture, trained and evaluated on a dataset of 39,531 samples. Their results demonstrate strong performance, underscoring the effectiveness of transformer-based models for entity extraction tasks. In a related direction, Huang et al. [12] proposed FinBERT, a domain-adapted language model tailored specifically for financial text. Their study illustrates that FinBERT captures domain-specific linguistic patterns more effectively than general-purpose models. It excels in summarizing contextual nuances found in financial documents, particularly in settings where the available training data is limited or includes specialized terminology rarely observed in general texts.

This study investigates how deep learning models with widely varying parameter sizes can be effectively applied to identify personal data in unstructured text through NER. To assess their capabilities, we conducted a systematic evaluation of a broad spectrum of language model architectures, including DistilBERT-base-uncased, DistilBERT-base-cased, BERT-base-uncased, BERT-base-cased, BERT-large-uncased, BERT-large-cased, ModernBERT-base, ModernBERT-large, nomic-BERT-2048, RoBERTa-base, DistilRoBERTa-base, RoBERTa-large, DeBERTa-v3-xsmall, DeBERTa-v3-small, and DeBERTa-v3-base. These models were compared using accuracy, precision, recall, and F1-score to examine how, and to what extent, certain SLMs can perform comparably to corresponding LLMs on a Personally Identifiable Information (PII)-specific dataset. Our findings reveal both the strengths and limitations of current approaches and demonstrate that SLMs can offer competitive performance, underscoring their viability for privacy-preserving information extraction tasks.

2. Materials and Methods

2.1. Dataset and Preprocessing

In this study, we employed the AI4Privacy PII 43 K dataset, a notable contribution to the intersection of privacy and artificial intelligence. Available through the ai4privacy/pii-masking-43 K repository [13], the dataset addresses the increasing need for robust personal data protection in AI-driven systems. It contains roughly 43,000 English-language samples generated through proprietary algorithms, enabling the creation of synthetic data that mitigates the risk of privacy breaches. Designed specifically for training AI models to detect and mask PII, the dataset spans 54 PII categories across 229 diverse use cases, including domains such as business and law. By providing a comprehensive foundation for data anonymization, this resource supports organizations that manage sensitive information, such as financial or medical institutions, in applying PII masking models to safely anonymize data prior to analysis, research, or data sharing. Each record in the dataset consists of six fields with corresponding text categories, as analytically described in Appendix A. Additionally, in Table A1 of Appendix A, we describe the records from the AI4Privacy PII 43 K dataset analytically.

2.2. Related Work and Comparative Analysis

Additionally, we performed a comparative analysis by identifying all existing works that referenced the NER-based AI4Privacy PII dataset used in our study. We executed two queries in Google Scholar. The first query searched for the exact dataset link on the Kaggle website ("https://www.kaggle.com/datasets/verracodeguacas/ai4privacy-pii") (accessed on Friday 10 October 2025), which returned only one result [14]. This paper was relevant because it reported evaluation metrics based on the AI4Privacy PII dataset, specifically the 43 K version, which is the same version used in our work, but their work [14] used only 42 out of the 56 PII classes by performing data cleaning. The second query searched for the dataset name ("AI4Privacy-PII"), which returned 33 results. Our Google Scholar search was conducted in October 2025.

From the 33 results, five were excluded for the following reasons:

- One result (the 16th) was a duplicate of the same work as in the first query [14].
- Three results (the 12th, 17th, and 26th) were citations within other publications and did not correspond to standalone works.
- Two results (the 25th and 29th) [15] represented the same work from different sources, so they were treated as a single publication.

From the remaining 28 works, none used the entire AI4Privacy PII 43 K dataset without data cleaning or data augmentation. These 28 works are categorized as follows:

- Five works in total used the 43 K version of the dataset: (i) three works used the 43 K version without demonstrating the same evaluation metrics as in our study (the 1st, 8th, and 21st results from the 33 results) [16–18], (ii) one work used only a fragment of the dataset (the 22nd result) (with the evaluation results of precision and recall) [19], and (iii) one work used the dataset while increasing contextual diversity by employing a synthetic PII extension (65 K version) without demonstrating the same evaluation metrics as in our study, which appeared twice in the search results (the 25th and 29th results), representing the same publication from different sources [15].
- Five works (the 2nd, 6th, 9th, 28th, and 33rd results) [20–24] mentioned the dataset only in their related work sections and did not use it in their experiments.
- Eight works (the 3rd, 5th, 7th, 14th, 18th, 19th, 23rd, and 32nd results) [25–32] used the 200 K version of the dataset.
- Seven works (the 4th, 11th, 13th, 15th, 20th, 27th, and 30th results) [33–39] used the 300 K version of the dataset.
- Two works (the 24th and 31st results) [40,41] used the 400 K version of the dataset.
- One work (the 10th result) [42] used both the 300 K and 500 K versions of the dataset.

From the two queries combined, only two works use the AI4Privacy PII 43 K dataset and provide evaluation metrics, but they apply data cleaning [14,19], unlike our work, which uses the entire dataset. The inclusion and exclusion process is illustrated in Figure 1. As described analytically above, among these two works, (i) one used 42 of the 56 PII classes [14] and (ii) one used only a fragment of the dataset [19].

2.3. Language Models Under Study and Experimental Setup

This work explores how deep learning models, with completely different numbers of parameters, can be effectively applied to detect personal data within unstructured text using NER techniques. Using the indices of accuracy, precision, recall, and F1-score, we evaluate the performance of various architectures by leveraging the following language models:

- BERT-large-uncased (2018): This larger configuration of BERT increases depth and parameter count, allowing it to model more subtle contextual relationships. Released with the original BERT paper, the uncased model remains a powerful baseline for many high-performance NLP tasks.
- BERT-large-cased (2018): Similarly introduced in 2018, this variant combines BERT-large's capacity with case-sensitive tokenization. It is particularly effective for entity-rich tasks, albeit with substantially higher computational cost.

- BERT-base-uncased (2018): BERT-base, introduced in 2018, set a new standard for contextualized language representations. The uncased version is suited to tasks where capitalization does not carry semantic importance but deep bidirectional context does.
- BERT-base-cased (2018): Published in the same foundational work, the cased version preserves case distinctions and, therefore, tends to perform better in tasks involving names, formal terminology, and other case-sensitive expressions.
- DistilBERT-base-uncased (2019): DistilBERT was introduced in 2019 as a compact alternative to BERT, designed to retain most of its capability while being faster and lighter. The uncased version ignores capitalization and performs well on broad text corpora where case is not critical.
- DistilBERT-base-cased (2019): Released alongside the uncased variant, the cased version keeps capitalization information, which can help identify names, acronyms, and structured entities. It offers a strong balance between efficiency and the ability to capture fine-grained textual cues.
- RoBERTa-base (2019): RoBERTa was introduced in 2019 as an improved, more robustly trained version of BERT. Its base model eliminates next sentence prediction and uses dynamic masking, leading to stronger performance across many downstream tasks.
- RoBERTa-large (2019): The large version, introduced alongside the base model in 2019, deepens the architecture to capture richer semantics. It provides excellent performance in detailed entity recognition tasks, though it requires considerably more compute.
- DistilRoBERTa-base (2019): Released soon after RoBERTa, this distilled model delivers much of RoBERTa's accuracy with a significantly smaller footprint. Its efficiency makes it ideal for real-time or resource-constrained applications.
- DeBERTa-v3-xsmall (2021): DeBERTa-v3 was released in 2021, incorporating disentangled attention and an improved masked language modeling objective. The xsmall version brings these benefits to a very compact model that still performs well on NER tasks.
- DeBERTa-v3-small (2021): Also introduced in 2021, this variant expands capacity while maintaining efficiency. It is well-suited for tasks requiring stronger contextual understanding without incurring heavy computational costs.
- DeBERTa-v3-base (2021): The base version of DeBERTa-v3 further leverages the improved architecture released in 2021, consistently outperforming similarly sized models in structured extraction tasks, including PII recognition.
- nomic-BERT-2048 (2023): Introduced in 2023, nomic-BERT was developed to handle significantly longer inputs, up to 2048 tokens. This capability makes it well-suited for processing long documents where important information may be spread far apart.
- ModernBERT-base (2023): ModernBERT appeared in 2023 with architectural refinements designed to speed up training and improve efficiency. The base version provides a balanced option for users who want a streamlined transformer without compromising too much on performance.
- ModernBERT-large (2023): Released in the same year, the large version increases the parameter count and depth, enabling more expressive modeling. While more demanding computationally, it typically offers stronger results in complex NER tasks.

Across these models, the underlying architectures share the same transformer foundation but differ in depth, parameter count, and design refinements that shape their performance profiles. The BERT and RoBERTa families rely on multi-layer bidirectional transformers that encode context from both directions, with the larger variants offering deeper stacks of attention layers and richer representational capacity. DistilBERT and DistilRoBERTa compress these architectures through knowledge distillation, reducing computation while attempting to retain the core contextual reasoning abilities. ModernBERT introduces architectural optimizations that streamline attention computations and accelerate processing, making it more efficient for large-scale inference. DeBERTa-v3 further departs from the traditional transformer structure by disentangling the attention mechanism and enhancing the masked language modeling objective, enabling the model to better separate content from positional information. Nomic-BERT adds another dimension by extending its context window to 2048 tokens, making it capable of handling longer sequences than conventional BERT-based models. While all these models operate within the transformer paradigm, their architectural nuances, ranging from depth and embedding size to optimized attention mechanisms, collectively shape how effectively they capture semantic relationships and identify entities in unstructured text.

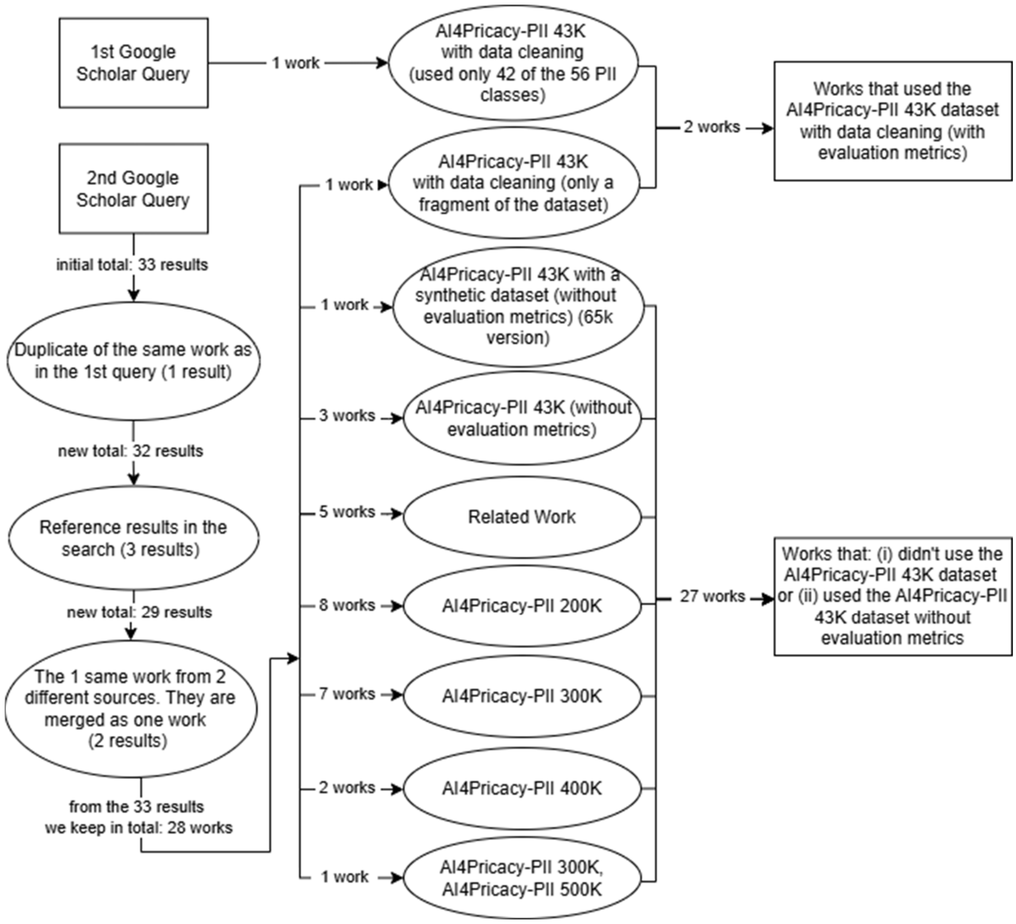

Figure 1. Comparative analysis based on the literature.

We studied the fine-tuning process of all these models. Table 1 contains the list of models with the year of publication, model category, number of parameters, and runtime.

All the experiments were conducted with a computer with an Intel i7 14,700 CPU and 64 GB of RAM, equipped with a GeForce RTX 5060 16 GB GDDR 7 GPU. Among all the training parameters, we chose epochs = 5, learning rate = 3×10^{-5}, batch_size = 32, and weight_decay = 0.01. The dataset, after sufficient normalization, was divided into training and test datasets in a ratio of 0.8/0.2, respectively. It is worth noting that during fine-tuning, the GPU memory consumption was at the upper limit (100%).

Table 1. Year of publication, category, runtime, and number of parameters of the studied models.

Model	Year	Category	Time (h:m)	Parameters
Bert-LARGE-uncased	2018	large	57:40	340 M
Bert-LARGE-cased	2018	large	62:20	340 M
Bert-base-uncased	2018	lightweight	16:20	110 M
Bert-base-cased	2018	lightweight	11:59	110 M
Distilbert-base-uncased	2019	lightweight	6:17	82 M
Distilbert-base-cased	2019	lightweight	6:19	82 M
RoBERTa-base	2019	lightweight	12:25	125 M
RoBERTa-large	2019	large	60:17	355 M
Distil-Roberta-base	2019	lightweight	10:41	82 M
Deberta-v3-xsmall	2021	lightweight	13:35	22 M
Deberta-v3-small	2021	lightweight	14:39	44 M
Deberta-v3-base	2021	lightweight	47:08	86 M
nomic-BERT-2048	2023	lightweight	1:19	137 M
ModernBERT-large	2023	large	106:54	395 M
ModernBERT-base	2023	lightweight	1:10	149 M

Although the field of LMs does not follow a strict or universally agreed-upon taxonomy for categorizing transformer models by parameter size, several influential works provide commonly adopted conventions. Foundational encoder models, such as BERT (~110 M parameters), are widely referred to as base models, with their larger counterparts (~340 M parameters) labeled large [43]. More recent studies and large model releases (e.g., GPT-3, PaLM) generally reserve the term LLMs for architectures that exceed billions of parameters [44,45]. Similarly, comprehensive surveys on pretrained model compression and scaling behavior indicate that models under ~50 M parameters are frequently treated as extra-small, those in the 50–100 M range as small, around 100–150 M as base, and 300–400 M as large [46–48].

In this study, we follow these widely used parameter ranges, as they align well with the BERT family architectures included in our experiments. Our evaluated models span from approximately 22 M to 395 M parameters, making it practical to group them into three meaningful categories: (i) lightweight (<150 M), (ii) medium (150–300 M), and (iii) large encoder variants (>300 M). This classification is consistent with community conventions and reflects the fact that models in the hundreds of millions of parameters are typically viewed as compact encoder/decoder architectures, not true LLMs [44,45]. The categories of all the evaluated models are shown in Table 1. By clearly defining the parameter categories used in our comparison, we aim to maintain transparency and ensure consistent terminology for readers and reviewers.

3. Results

In this section, we present the comparison of our work with other similar studies in terms of the performance metrics of accuracy, precision, f1-score, and loss, and then we discuss the results. Additionally, we compare our work with other studies conducted on the same or similar datasets, as demonstrated in Table 2.

In Tables 3–5, the change in performance metrics per epoch for the best models of RoBERTa-large, nomic-BERT-2048, and RoBERTa-base are presented, respectively.

https://doi.org/10.3390/app16073332

Table 2. Performance indices of our fine-tuned models and comparison with other works.

Works	Models	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	Loss (%)
	Bert-LARGE-uncased	96.9	92.19	94.65	93.41	7.43
	Bert-LARGE-cased	96.88	91.98	94.56	93.25	8.88
	Bert-base-uncased	96.82	91.52	94.15	92.82	7.94
	Bert-base-cased	96.84	90.7	94.02	92.33	7.73
	Distilbert-base-uncased	96.54	90.44	93.31	91.85	8.21
	Distilbert-base-cased	96.76	89.95	93.41	91.64	8.12
	RoBERTa-base	96.94	92.10	94.37	93.11	7.76
	RoBERTa-large	97.19	92.73	94.94	93.82	7.47
	Distil-Roberta-base	96.76	90.45	93.27	91.84	8.18
	nomic-BERT-2048	97.03	92.61	94.63	93.61	7.46
	ModernBERT-base	96.75	85.2	91.23	88.44	8.07
	ModernBERT-large	96.89	88.35	93.49	90.85	7.27
Areskoug, T., & Mörk, V. (2025) [14] (with data cleaning, they only have 42 of the 56 PII classes)	BiLSTM	-	90.5	89.5	90.0	-
	BiLSTM-CRF	-	92.6	93.9	93.2	-
	BERT	-	95.8	95.7	95.8	-
	BERT-CRF	-	95.8	95.9	95.9	-
	DistillBERT	-	95.7	96.1	95.9	-
	DistillBERT-CRF	-	95.7	96.4	96.1	-
	DeBERTa	-	96.5	97.0	96.7	-
	DeBERTa-CRF	-	96.8	96.8	96.8	-
	ModernBERT	-	96.1	95.8	96.0	-
	ModernBERT-CRF	-	96.3	96.5	96.4	-
Zhou et al. (2025) [19] (with data cleaning, they used a fragment of the dataset)	Rescriber-GPT-4o	-	94	88	-	-
	Rescriber-Llama3-8B	-	94	63	-	-
	Presidio	-	69	68	-	-

Table 3. Changes in performance metrics per epoch are for the model RoBERTa-large.

Epochs	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	Loss (%)
1	95.16	89.27	92.06	90.64	9.65
2	96.61	89.39	93.35	91.33	7.95
3	97.03	92.22	94.46	93.32	7.57
4	97.19	92.21	94.92	93.54	7.47
5	96.90	92.73	94.94	93.82	8.94

Table 4. Changes in performance metrics per epoch are for the model nomic-BERT-2048.

Epochs	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	Loss (%)
1	95.42	85.23	89.86	87.48	9.82
2	96.64	90.64	93.12	91.86	8.35
3	96.95	91.75	94.17	92.94	7.46
4	97.03	92.53	94.62	93.56	7.82
5	96.85	92.61	94.63	93.61	8.56

Table 5. Changes in performance metrics per epoch are for the model RoBERTa-base.

Epochs	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	Loss (%)
1	95.31	85.38	89.87	87.57	10.39
2	96.56	89.40	92.15	90.75	8.52
3	96.86	91.94	93.89	92.90	7.76
4	96.94	92.10	94.30	93.10	7.80
5	96.78	91.89	94.37	93.11	8.17

The comparison among all models we studied on the basis of F1-score is illustrated in Figure 2.

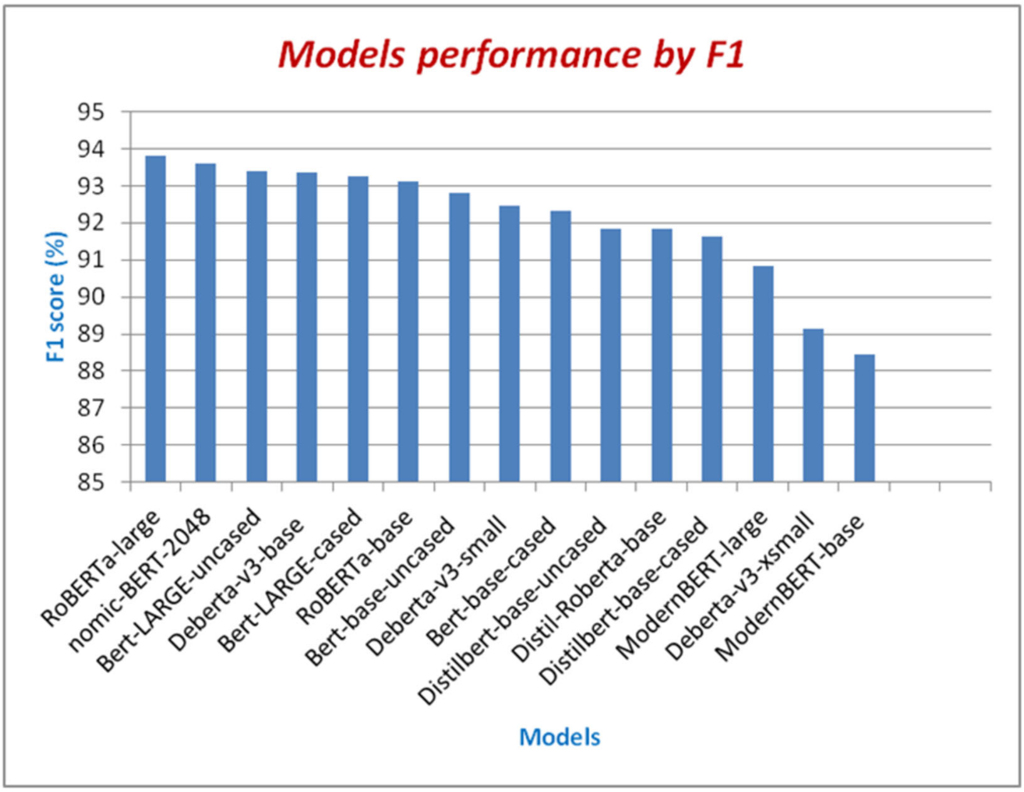

Figure 2. Comparison among all models on the basis of F1-score.

The comparison among all models we studied on the basis of the accuracy index is illustrated in Figure 3.

Additionally, Table 6 demonstrates each model's evaluation samples and steps per second.

Our experiments show that some SLMs compete equally with some corresponding LLMs, based on the specific PII dataset, thus enhancing personal data detection, which is of paramount importance in financial applications. The evaluation of fifteen transformer-based architectures across accuracy, precision, recall, F1-score, training loss, and computational cost reveals several notable patterns about model performance on our dataset. Overall, all models achieve high accuracy ($\geq$96%), indicating that the task is well-aligned with transformer-based language understanding. However, the subtler distinctions across precision, recall, and F1-score highlight how architectural choices and model size influence reliability and generalization. Across the full set of models, the highest F1-scores are obtained by RoBERTa-large (93.82%), nomic-BERT-2048 (93.61%), BERT-LARGE-uncased (93.41%), and DeBERTa-v3-base (93.36%). These systems consistently balance strong precision with high recall, resulting in the most robust overall performance. As expected, these models also come with substantially higher parameter counts, ranging from 137 M to 355 M, and significantly longer training times. For example, ModernBERT-large required over 106 h of training, while BERT-LARGE-cased exceeded 62 h.

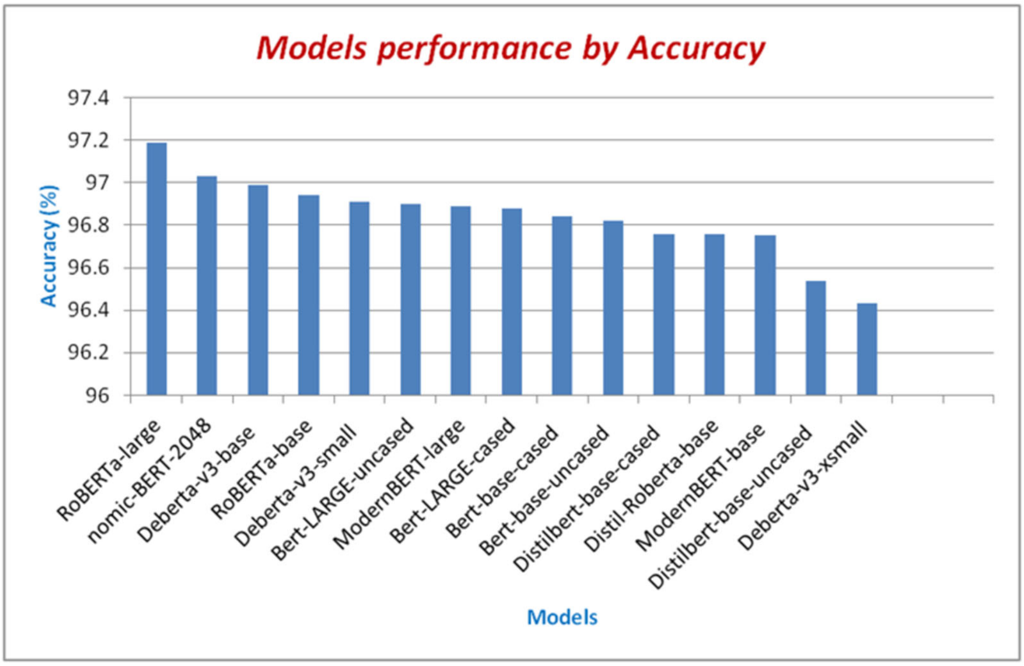

Figure 3. Comparison among all models on the basis of the accuracy index.

Table 6. Each model's evaluation samples and steps per second.

Model	Evaluation Samples per Second	Evaluation Steps per Second
Bert-LARGE-uncased	5.013	0.313
Bert-LARGE-cased	5.512	0.345
Bert-base-uncased	7.479	0.468
Bert-base-cased	8.687	0.543
Distilbert-base-uncased	20.886	1.306
Distilbert-base-cased	21.047	1.316
RoBERTa-base	8.365	0.54
RoBERTa-large	8.298	0.519
Distil-Roberta-base	9.988	0.624
Deberta-v3-xsmall	6.204	0.388
Deberta-v3-small	10.679	0.668
Deberta-v3-base	8.031	0.502
nomic-BERT-2048	143.715	8.985
ModernBERT-large	1.381	0.086
ModernBERT-base	128.04	8.005

To contextualize the performance of our models, we compare our results against those reported by Areskoug and Mörk (2025) [14], who evaluated a range of architectures, including BiLSTM variants, BERT-based models, DistilBERT, DeBERTa, and ModernBERT on a similar personal data detection task. Across our experiments, the highest F1-scores are obtained by RoBERTa-large (93.82%), nomic-BERT-2048 (93.61%), DeBERTa-v3-base (93.36%), and BERT-LARGE-uncased (93.41%). A closer inspection of the ranking positions further clarifies the comparative behavior of large and small models. While the single best-performing model in both accuracy and F1-score is a large architecture (RoBERTa-large), the remaining large models do not consistently dominate the top positions. Specifically, the four large models

(RoBERTa-large, BERT-LARGE-uncased, ModernBERT-large, and BERT-LARGE-cased) are ranked first, sixth, seventh, and eighth in terms of accuracy and first, third, fifth, and 13th in terms of F1-score, respectively. In contrast, the positions immediately following the top-ranked model are largely occupied by SLMs. For accuracy, SLMs hold positions 2 through 5, while for F1-score, they occupy positions 2 and 4. This ranking-based comparison shows that, aside from the top result, smaller and mid-sized models consistently achieve performance comparable to, or better than, most large architectures, providing direct evidence that SLMs are strong competitors to LLMs on this PII-specific NER task. These results place our best-performing models very close to, though slightly below, the top values reported by Areskoug and Mörk [14]. For example, their strongest models, DeBERTa and DeBERTa-CRF, reach F1-scores of 96.7% and 96.8%, respectively. Similarly, their BERT, BERT-CRF, and DistilBERT-CRF models achieve F1-scores around 95.8–96.1%. Although their highest scores exceed ours by a small margin, it is important to consider the broader context of model evaluation, reporting completeness, and computational transparency. Additionally, it is worth noting that their work [14] used only 42 of the 56 PII classes, whereas we used all 56.

Zhou et al. [19] further highlight the importance of evaluation conditions when comparing reported performance. Their study relies on substantial data cleaning and, notably, on using only a fragment of the original dataset, which significantly simplifies the detection task. While their GPT-4o-based Rescriber achieves a precision of 94%, its recall drops to 88%, indicating that a considerable portion of sensitive entities remains undetected. This limitation is even more pronounced for the Rescriber-Llama3-8B model, where recall falls sharply to 63%, despite similar precision. Such an imbalance suggests that these systems may be overly conservative, prioritizing precision at the expense of comprehensive coverage. In contrast, our models consistently maintain both high precision and high recall across all 56 PII classes, resulting in F1-scores above 93% for several architectures. Importantly, this performance is achieved without data cleaning or class reduction, making our results more representative of real-world financial scenarios.

Overall, this places our work at a clear competitive advantage. By evaluating a broad range of models under identical and transparent conditions, and using the full dataset with all 56 PII classes, we provide a more realistic and demanding benchmark than prior studies. Our results show that strong and well-balanced PII detection does not require extremely large or costly models, nor heavy data filtering, but can be achieved with carefully selected transformer architectures that generalize well across diverse entity types.

4. Discussion

While the large architectures generally lead the rankings, what stands out is how closely several smaller models trail behind. For instance, DeBERTa-v3-small (44 M parameters) achieves an F1-score of 92.46%, almost identical to BERT-base-cased (92.33%) despite having less than half the parameters and a run time of only 14 h compared to 12 h for BERT-base, remarkably efficient given its size. DistilBERT-base-uncased and DistilBERT-base-cased (82 M parameters) maintain F1-scores of around 91.6–91.8%, only about two percentage points behind the best-performing large models, while the run time is just over six hours. DistilRoBERTa-base also performs strongly, with an F1-score of 91.84% and a comparatively short training time of 10 h. These findings reinforce that, although large models deliver marginally higher performance, the gap is much smaller than their computational cost suggests.

Furthermore, we have the following architecture-based observations: RoBERTa variants consistently perform near the top, with the large variant achieving the best overall F1-score (93.82%), confirming RoBERTa's well-known strengths in masked language model pretraining. BERT-based models show predictable scaling behavior: performance improves from DistilBERT to BERT-base, and from BERT-base to BERT-large, though the gains shrink as the model capacity increases. DeBERTa-v3 models, especially the base version, show strong behavior across all metrics, benefiting from their enhanced disentangled attention and improved vocabulary embedding structure. ModernBERT, while extremely fast to train, underperforms slightly in precision and recall relative to the other architectures. Nevertheless, its speed, particularly the base version, which finishes in just over an hour, makes it promising for scenarios requiring rapid experimentation.

Additionally, we notice that run time varies enormously between models, from 1 h and 10 min for ModernBERT-base to nearly 58–62 h for BERT-LARGE variants. When combining performance and time cost, several models achieve an excellent balance. DistilBERT models achieve competitive results at a fraction of the run time. Nomic-BERT-2048 is especially efficient, offering the third-highest F1-score while the run time lasts for only about 80 min, making it an exceptional speed–performance trade-off. DeBERTa-v3-small also delivers near state-of-the-art results at moderate computational cost. These patterns support the broader conclusion that small and mid-size models can rival large ones on specialized tasks, especially when the dataset is well-structured and domain-specific.

5. Conclusions

Our study demonstrates that certain SLMs can perform as well as their larger counterparts (LLMs) when evaluated on a specific PII dataset, highlighting their potential for enhancing personal data detection, an especially critical task in financial applications. Additionally, we introduced a novel architecture that leverages an optimized transformer fine-tuning strategy to improve PII recognition across varied contexts. Through an extensive comparative analysis using the AI4Privacy PII 43 K dataset, we benchmarked our approach against all relevant publicly available methods, providing a comprehensive assessment of current solutions. Overall, the results clearly support our conclusion that SLMs are strong competitors to LLMs. While the highest accuracy and F1-score are achieved by a large model, SLMs consistently follow in the top rankings, holding positions 2–5 for accuracy and positions 2 and 4 for F1-score. This close performance gap highlights that smaller models can deliver results comparable to much larger architectures, reinforcing their practical value for privacy-preserving NER in real-world financial settings. The results not only underscore the strengths and weaknesses of existing approaches but also confirm the effectiveness of SLMs in addressing the challenges of privacy-preserving information extraction.

For future work, we plan to conduct a per-class confusion matrix for all the models we examined, as well as to expand our study to include additional SLMs and LLMs, further enriching the comparative evaluation and deepening our understanding of their performance dynamics.

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Abbreviations

The following abbreviations are used in this manuscript:

NER	Named Entity Recognition
LMs	Language Models
SLMs	Small Language Models
LLMs	Large Language Models
PII	Personally Identifiable Information
TTPs	Tactics, Techniques, and Procedures
RAG	Retrieval Augmented Generation
NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from Transformers

Appendix A

Each record in the AI4Privacy PII 43 K dataset consists of six fields with corresponding text categories:

1. Masked text: contains the original text where PII words were masked;
2. Unmasked text: which is the original text;
3. Privacy mask: the PII classes that belong to each sentence;
4. Span labels: the spanning range of each PII class;
5. BIO labels: where we have the labels of the PII classes;
6. Tokenized text: each sentence is split into its respective tokens.

Let us assume we have the class "phone_IMEI_number" (where IMEI stands for International Mobile Equipment Identity), where:

1. B is the beginning of the PII text: B-phone_IMEI_number;
2. I is within the PII text: I-phone_IMEI_number;
3. O is the nOn PII class of the text: O.

The dataset consists of PII classes (e.g., ACCOUNTNUMBER, CREDITCARDISSUER, BITCOINADDRESS, CREDITCARDCVV, CURRENCYCODE, PASSWORD, PHONEIMEI, SNN, USERNAME, EMAIL).

Examples from the record set that we used [13] are shown below in Table A1.

Table A1. Records from the AI4Privacy PII 43 K dataset.

Masked Text	Unmasked Text	Privacy Mask	Span Labels	Tokenized Text	BIO Labels
A student's assessment was found on a device bearing IMEI: [PHONEIMEI_1]. The document falls under the various topics discussed in our [JOBAREA_1] curriculum. Can you please collect it?	A student's assessment was found on a device bearing IMEI: 06-184755-866851-3. The document falls under the various topics discussed in our optimization curriculum. Can you please collect it?	{'[PHONEIMEI_1]': '06-184755-866851-3', '[JOBAREA_1]': 'Optimization'}	[[0, 57, 'O'], [57, 75, 'PHONEIMEI_1'], [75, 138, 'O'], [138, 150, 'JOBAREA_1'], [150, 189, 'O']]	["O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "B-PHONEIMEI", "I-PHONEIMEI", "I-PHONEIMEI", "I-PHONEIMEI", "I-PHONEIMEI", "I-PHONEIMEI", "I-PHONEIMEI", "I-PHONEIMEI", "I-PHONEIMEI", "I-PHONEIMEI", "I-PHONEIMEI", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "B-JOBAREA", "O", "O", "O", "O", "O", "O", "O", "O"]	["a", "student", "'", "s", "assessment", "was", "found", "on", "device", "bearing", "im", "##ei", ":", "06", "-", "1847", "##55", "-", "86", "##6", "##85", "##1", "-", "3", ".", "the", "document", "falls", "under", "the", "various", "topics", "discussed", "in", "our", "optimization", "curriculum", ".", "can", "you", "please", "collect", "it", "?"]}

Table A1. *Cont.*

Masked Text	Unmasked Text	Privacy Mask	Span Labels	Tokenized Text	BIO Labels
Dear [FIRSTNAME_1], as per our records, your license [VEHICLEVIN_1] is still registered in our records for access to the educational tools. Please provide feedback on its operability.	Dear Omer, as per our records, your license 78B5R2MVFAHJ48500 is still registered in our records for access to the educational tools. Please provide feedback on its operability.	{'[FIRSTNAME_1]': 'Omer', '[VEHICLEVIN_1]': '78B5R2MVFAHJ48500'}	[[0, 5, 'O'], [5, 9, 'FIRSTNAME_1'], [9, 44, 'O'], [44, 61, 'VEHICLEVIN_1'], [61, 170, 'O']]	bio_labels": ["O", "B-FIRSTNAME", "I-FIRSTNAME", "O", "O", "O", "O", "O", "O", "O", "O", "B-VEHICLEVIN", "I-VEHICLEVIN", "I-VEHICLEVIN", "I-VEHICLEVIN", "I-VEHICLEVIN", "I-VEHICLEVIN", "I-VEHICLEVIN", "I-VEHICLEVIN", "I-VEHICLEVIN", "I-VEHICLEVIN", "I-VEHICLEVIN", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O"]	tokenised_text": ["dear", "om", "##er", ",", "as", "per", "our", "records", ",", "your", "license", "78", "##b", "##5", "##r", "##2", "##m", "##v", "##fa", "##h", "##j", "##48", "##500", "is", "still", "registered", "in", "our", "records", "for", "access", "to", "the", "educational", "tools", ".", "please", "feedback", "on", "it", "'", "s", "opera", "##bility", "."]}
[FIRSTNAME_1], could you please share your recommendations about a vegetarian diet for [AGE_1] old [GENDER_1] with [HEIGHT_1]?	Kattie, could you please share your recommendations about a vegetarian diet for a 72-year-old intersex person who is 158 centimeters tall?	{'[FIRSTNAME_1]': 'Kattie', '[AGE_1]': '72', '[GENDER_1]': 'Intersex person', '[HEIGHT_1]': '158 centimeters'}	[[0, 6, 'FIRSTNAME_1'], [6, 75, 'O'], [75, 77, 'AGE_1'], [77, 82, 'O'], [82, 97, 'GENDER_1'], [97, 103, 'O'], [103, 117, 'HEIGHT_1'], [117, 118, 'O']]	["B-FIRSTNAME", "I-FIRSTNAME", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "B-AGE", "O", "B-GENDER", "I-GENDER", "I-GENDER", "I-GENDER", "O", "B-HEIGHT", "I-HEIGHT", "I-HEIGHT", "O"]	["kat", "##tie", "could", "you", "please", "share", "your", "rec", "##om", "##nda", "##tions", "about", "vegetarian", "diet", "for", "72", "old", "inter", "##se", "##x", "person", "with", "158", "##cent", "##imeters", "?"]}
Emergency supplies in [BUILDINGNUMBER_1] need a refill. Use [MASKEDNUMBER_1] to pay for them."	Emergency supplies in 16356 need a refill. Use 5890724654311332 to pay for them.	{'[BUILDINGNUMBER_1]': '16356', '[MASKEDNUMBER_1]': '5890724654311332'}	[[0, 22, 'O'], [22, 27, 'BUILDINGNUMBER_1'], [27, 47, 'O'], [47, 63, 'MASKEDNUMBER_1'], [63, 80, 'O']]	["O", "O", "O", "B-BUILDINGNUMBER", "I-BUILDINGNUMBER", "O", "O", "O", "O", "O", "O", "B-MASKEDNUMBER", "I-MASKEDNUMBER", "I-MASKEDNUMBER", "I-MASKEDNUMBER", "I-MASKEDNUMBER", "I-MASKEDNUMBER", "I-MASKEDNUMBER", "I-MASKEDNUMBER", "I-MASKEDNUMBER", "O", "O", "O", "O", "O"]	["emergency", "supplies", "in", "1635", "##6", "need", "a", "ref", "##ill", ".", "use", "58", "##90", "##7", "##24", "##65", "##43", "##11", "##33", "##2", "to", "pay", "for", "them", "."]}

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